

# Travis Brown

browntravis760@gmail.com • 518-796-8703 • github.com/tmb5932 • www.travismbrown.com

## OBJECTIVE

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Seeking co-op in embedded software engineering from January 2026 to August 2026.

## EDUCATION

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Rochester Institute of Technology

Bachelor of Science in Computer Science, Minor in Criminal Justice

GPA: 3.94

Rochester, NY

Expected: May 2027

## SKILLS

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**Languages:** C, C++, Python, Java, JavaScript, Rust, SQL, MIPS Assembly

**Embedded:** Logic Analyzer, OpenOCD, STM32 MCUs, SPI, I2C, UART, PWM, ADC

**Tools/Environments:** Git, GitHub, GitLab, Linux, MacOS, PostgreSQL

## WORK EXPERIENCE

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### QA Automation Intern

Lockheed Martin

May 2025 – August 2025

- Produced a Python-based automation tool that streamlined static code review, reducing QA review time by 75% (from 20 mins to 5 mins).
- Automated over 60% of manual software validation checks, significantly reducing QA workload and improving team efficiency.
- Collaborated with QA engineers to identify and resolve software issues, improving release reliability and reducing defects.
- Built and maintained CI/CD pipelines using GitLab CI, contributing to faster and more reliable testing cycles.
- Improved scalability and maintainability of Python automation scripts by modularizing code and implementing unit tests.

### Digital Content Assistant

Spire Business Inc

June 2024 – April 2025

- Created SEO-optimized web pages with HTML/CSS per owner's specifications.
- Resolved common computer issues, such as slow performance, application errors, and login difficulties, ensuring minimal downtime.

## PROJECTS

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### Handheld Chip-8 Emulator Console

Personal Project

June 2025 – Present

- Developed a Chip-8 emulator in Rust for a Raspberry Pi, with display, input, and sound using GPIO and SPI communication.
- Designed a 4x4 matrix keypad PCB in KiCad, including schematic and layout, and manually soldered key switches, caps, and diodes.
- Created a custom SPI display driver for an SSD1309 OLED screen, avoiding external graphics libraries for low-level control.

### Real-Time GPS Tracking with Wireless Communication

RIT Electric Vehicle Team

January 2025 – Present

- Enabled tracking of vehicle position, speed, and elevation changes during testing, supporting performance evaluation and driver feedback.
- Programmed low-level hardware drivers in C on the ESP32 using the ESP-IDF framework.
- Acquired real-time GPS data using the NMEA 0183 protocol from the TESEO-LIV3F GNSS module.
- Expanding the system with LoRa communication to transmit the GPS data to a remote computer for real-time telemetry analysis.

### STM32 F4 Peripheral Support

RIT Electric Vehicle Team

December 2023 – November 2024

- Implemented driver support for PWM, ADC, SPI, and timers on the STM32F4 microcontroller family, working with the STM32 Hardware Abstraction Layer.
- Worked with a multidisciplinary team to develop C/C++ firmware for custom-designed PCBs.
- Leveraged STM32's CubeMX and CubeIDE to streamline configuration and integration with EVT's hardware-agnostic firmware API.
- Applied information from datasheets and reference manuals to develop reliable peripheral drivers.